

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

SIF List & SIL Determination (LOPA)

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1. Purpose

This report lists the safety instrumented functions (SIFs) of the CDU / VDU, summarises the SIL determination by layer of protection analysis (LOPA), and documents the trip set-point basis used in the cause & effect matrix CFU-000-IC-CE-001. It is the FEED input to the Safety Requirements Specification (SRS, IEC 61511-1 clause 10).

2. Method

- Scenarios from the preliminary HAZOP (CFU-000-PR-RPT-002) and the SIF list in cfu/control_loops.py.
- Tolerable / target mitigated event likelihood (TMEL) by consequence category: C5 multiple fatalities 1e-6 /yr, C4 single fatality 1e-5 /yr, C3 serious injury / major environmental or asset 1e-4 /yr, C2 1e-3 /yr.
- Initiating event frequencies and IPL PFDs are CCPS typical values: BPCS loop failure 0.1 /yr; operator response to an independent alarm with ≥ 10 min available 0.1; PSV 0.01. Conditional modifiers (ignition, occupancy) are applied explicitly.
- Required risk reduction $RRF = (IEF \times \text{prod(IPL PFD)} \times \text{prod(CM)}) / \text{TMEL}$; SIL 1: $10 < RRF \leq 100$, SIL 2: $100 < RRF \leq 1000$, SIL 3: $1000 < RRF \leq 10000$.
- Manual ESD (SIF-901) is assigned SIL 2 by company practice (no LOPA scenario).

3. SIF list

SIF	Function	Initiators	Voting	Final elements	SIL	Response	Proof test
SIF-101	H-101 low-low pass flow (any pass) -> trip fuel	FT-1011..1018 (2oo3 each pass)	2oo3	XV-1021/1022 FG SSOV, XV-1026 pilot	SIL 2	2 s	12 months
SIF-102	H-101 low-low fuel gas pressure -> trip fuel	PT-1027A/B/C 2oo3	2oo3	XV-1021/1022	SIL 1	2 s	24 months
SIF-103	H-101 high-high fuel gas pressure -> trip fuel	PT-1027A/B/C 2oo3	2oo3	XV-1021/1022	SIL 1	2 s	24 months
SIF-104	H-101 loss of flame (all burners) -> trip fuel	BS-1028 (scanners)	per burner / all	XV-1021/1022/1026	SIL 2	4 s (FFRT)	12 months
SIF-105	H-101 high-high arch pressure / loss of ID fan -> trip	PT-1029A/B/C	2oo3	XV-1021/1022	SIL 1	3 s	24 months
SIF-106	H-101 high-high COT -> trip fuel	TT-1020A/B/C 2oo3	2oo3	XV-1021/1022	SIL 1	5 s	24 months
SIF-107	D-101A/B low-low interface (grid short) -> trip transformers	LT-1007B / LT-1008B	1oo1	Desalter transformer CB	SIL 1	5 s	24 months
SIF-108	C-101 high-high bottom level -> stop P-101 / close FV-1001	LT-1082B	1oo1	XV-1001 + P-101 trip	SIL 1	10 s	24 months
SIF-109	D-105 low-low level -> close LPG product XV (gas blow-by to LPG treating)	LT-1092B	1oo1	XV-1093	SIL 1	5 s	24 months

SIF	Function	Initiators	Voting	Final elements	SIL	Response	Proof test
SIF-110	C-105 high-high pressure -> cut reboiler steam	PT-1091B	1oo1	XV-1096	SIL 1	5 s	24 months
SIF-201	H-201 low-low pass flow -> trip fuel	FT-2001..2004	2oo3	XV-2006A/B	SIL 2	2 s	12 months
SIF-202	H-201 low-low FG pressure / flame failure -> trip	PT-2009A/B/C, BS-2008	per burner / all	XV-2006A/B	SIL 1	4 s (FFRT)	24 months
SIF-203	C-201 high-high bottom level -> trip FV-1083 / P-112	LT-2024B	1oo1	XV-1083	SIL 1	10 s	24 months
SIF-204	Pump P-112/P-204 hot-pump emergency isolation (fire)	Manual HS / fire detection	1oo2	EIV-1121/EIV-2041 (ROSOV)	SIL 1	30 s (valve)	24 months
SIF-901	Unit emergency shutdown (ESD-1, manual)	HS-9000 (CCR + field)	1oo2	All unit trip valves, heater trips	SIL 2	2 s	12 months

4. LOPA / SIL determination

4.1 Scenarios and initiating events

SIF	Hazard scenario	Initiating event (IEF /yr)	Consequence (TMEL /yr)
SIF-101	Loss of pass flow (FV fails closed, coking, charge loss) -> tube overheating / rupture, firebox fire	BPCS loop failure / charge pump trip (0.1)	C4 single fatality (1e-05)
SIF-102	Flame-out on low FG pressure -> fuel accumulation, re-ignition explosion	PV-1021 fails closed / FG supply loss (0.1)	C4 single fatality (1e-05)
SIF-103	High FG pressure -> flame lift-off / unstable flame -> flame-out, explosion	PV-1021 fails open (0.1)	C4 single fatality (1e-05)
SIF-104	Loss of flame with fuel flowing -> firebox explosion	Burner instability / air-fuel upset (0.1)	C4 single fatality (1e-05)
SIF-105	Positive firebox pressure (ID fan trip / damper closed) -> flue gas / flame release	ID fan trip or damper failure (0.2)	C3 serious injury (1e-04)
SIF-106	High COT -> coking / tube overheating, transfer-line overpressure	TIC-1020 / PIC-1021 failure (0.1)	C3 serious injury / major asset (1e-04)
SIF-107	Low interface -> water on electrodes, grid short / arcing, vapour generation, fire	LIC-1007/1008 failure (0.1)	C3 serious injury (1e-04)
SIF-108	C-101 overfill -> liquid into overhead / PSV liquid relief, flare carry-over	LIC-1082 / FV-1083 failure, P-112 trip (0.1)	C3 major environmental (1e-04)
SIF-109	D-105 loss of level -> gas blow-by into LPG treating / rundown, overpressure	LIC-1092 / FV-1093 fails open (0.1)	C3 serious injury (1e-04)
SIF-110	C-105 overpressure on reboiler upset / loss of cooling -> PSV-1005 lift, LPG release	FIC-1096 failure / A-106 fan loss (0.2)	C3 serious injury (1e-04)

SIF	Hazard scenario	Initiating event (IEF /yr)	Consequence (TMEL /yr)
SIF-201	H-201 loss of pass flow -> coking, tube rupture, fire	FIC-2001..2004 failure / P-112 trip (0.1)	C4 single fatality (1e-05)
SIF-202	H-201 low FG pressure / flame failure -> fuel accumulation, explosion	PV-2006 failure / burner instability (0.1)	C4 single fatality (1e-05)
SIF-203	C-201 overflow -> liquid into flash zone / wash bed, loss of vacuum, overpressure	LIC-2024 / FV-2025 failure, P-204 trip (0.1)	C3 major asset (1e-04)
SIF-204	Fire at hot pump (> AIT) -> escalation; isolate inventory of C-101 / C-201 bottoms	Pump seal failure with ignition (0.01)	C4 single fatality (1e-05)
SIF-901	Major emergency - unit-wide isolation (manual ESD)	Operator-initiated (escalation) (-)	C4 single fatality (1e-05)

4.2 Independent protection layers, conditional modifiers and required risk reduction

SIF	IPLs (PFD)	Conditional modifiers	MEL w/o SIF	RRF	SIL (LOPA)	SIL (SRS)
SIF-101	Low-flow alarm FAL + operator (>= 10 min) (0.1)	Occupancy (heater area) (0.25)	2.5e-03	250	SIL 2	SIL 2
SIF-102	Low-pressure alarm + operator (0.1)	Probability of delayed ignition (0.1); Occupancy (0.25)	2.5e-04	25	SIL 1	SIL 1
SIF-103	High-pressure alarm + operator (0.1)	Probability of flame-out given HH (0.1); Occupancy (0.25)	2.5e-04	25	SIL 1	SIL 1
SIF-104	Operator observation (no credit - too fast) (1)	Probability of explosive accumulation (0.1); Probability of ignition (0.5); Occupancy (0.25)	1.3e-03	125	SIL 2	SIL 2
SIF-105	Draft alarm + operator (0.1)	Occupancy (platform) (0.1)	2.0e-03	20	SIL 1	SIL 1
SIF-106	High-temperature alarm + operator (0.1)	Probability of tube rupture (0.5)	5.0e-03	50	SIL 1	SIL 1
SIF-107	Interface alarm + operator (0.1)	Probability of ignition (0.3)	3.0e-03	30	SIL 1	SIL 1
SIF-108	High-level alarm + operator (0.1)	Probability PSV relieves liquid (0.5)	5.0e-03	50	SIL 1	SIL 1
SIF-109	Low-level alarm + operator (0.1)	Probability of downstream LOPC (0.3)	3.0e-03	30	SIL 1	SIL 1
SIF-110	PSV-1005 (sized for case) (0.01)	Probability of ignition of flare release (1)	2.0e-03	20	SIL 1	SIL 1
SIF-201	Low-flow alarm + operator (0.1)	Occupancy (0.25)	2.5e-03	250	SIL 2	SIL 2
SIF-202	Alarm + operator (0.1)	Probability of delayed ignition (0.1); Occupancy (0.25)	2.5e-04	25	SIL 1	SIL 1
SIF-203	High-level alarm + operator (0.1)	Probability of damage (0.5)	5.0e-03	50	SIL 1	SIL 1
SIF-204	Fire-fighting (OSBL response) (1)	Probability of escalation within 15 min (0.1); Occupancy (0.5)	5.0e-04	50	SIL 1	SIL 1

SIF	IPLs (PFD)	Conditional modifiers	MEL w/o SIF	RRF	SIL (LOPA)	SIL (SRS)
SIF-901	-	-	-	-	SIL 2 (company standard, manual ESD - LOPA n/a)	SIL 2

Result: 15 SIFs assessed; 15 LOPA results are consistent with the SIL in the SIF list. SIL distribution: SIL 1: 11, SIL 2: 4. No SIL 3 function is required; the logic solver is nevertheless SIL 3 capable to allow future changes without hardware replacement.

5. Trip set-point basis

- Pass flow LL = 40 % of design pass flow (below the 50 % turndown minimum with 10 % margin).
- COT HH = design COT + 15 °C (H-101) / + 12 °C (H-201); high alarm at + 8 °C (+ 6 °C).
- Burner FG pressure LL / HH per burner vendor stability curve (FEED: 0.15 / 2.2 barg with 2.0 barg max normal at design firing).
- Arch pressure HH = +2.5 mmH₂O (normal -2.5 mmH₂O draft).
- Vessel pressure HH = min(94 % of PSV set, set - 0.5 bar) - keeps >= 6 % margin to PSV lift.
- Level HH = 85 % of transmitter span (above HLA at 75 %); LL = 15 % (below LLA at 25 %).

Initiator	SIF	Description	Voting	Normal	Trip	Response
FZLL-1011	SIF-101	H-101 pass 1 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1012	SIF-101	H-101 pass 2 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1013	SIF-101	H-101 pass 3 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1014	SIF-101	H-101 pass 4 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1015	SIF-101	H-101 pass 5 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1016	SIF-101	H-101 pass 6 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1017	SIF-101	H-101 pass 7 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
FZLL-1018	SIF-101	H-101 pass 8 flow low-low	2oo3	71.1 t/h	28.5 t/h	2 s
PZLL-1027	SIF-102	H-101 burner FG pressure low-low	2oo3	1.0-2.0 barg	0.15 barg	2 s
PZHH-1027	SIF-103	H-101 burner FG pressure high-high	2oo3	1.0-2.0 barg	2.2 barg	2 s
BZLL-1028	SIF-104	H-101 loss of flame (all burners)	per burner / all	flame on	flame off	4 s (FFRT)
PZHH-1029	SIF-105	H-101 arch pressure high-high / ID fan loss	2oo3	-2.5 mmH ₂ O	+2.5 mmH ₂ O	3 s
TZHH-1020	SIF-106	H-101 COT high-high	2oo3	366 °C	381 °C	5 s
LZLL-1007	SIF-107	D-101A interface low-low (grid short)	1oo1	50 %	15 %	5 s
LZLL-1008	SIF-107	D-101B interface low-low (grid short)	1oo1	50 %	15 %	5 s
LZHH-1082	SIF-108	C-101 bottom level high-high	1oo1	50 %	85 %	10 s

Initiator	SIF	Description	Voting	Normal	Trip	Response
LZLL-1092	SIF-109	D-105 level low-low (gas blow-by to LPG)	1oo1	50 %	15 %	5 s
PZHH-1091	SIF-110	C-105 pressure high-high	1oo1	10.8 barg	12.2 barg	5 s
FZLL-2001	SIF-201	H-201 pass 1 flow low-low	2oo3	64.5 t/h	25.8 t/h	2 s
FZLL-2002	SIF-201	H-201 pass 2 flow low-low	2oo3	64.5 t/h	25.8 t/h	2 s
FZLL-2003	SIF-201	H-201 pass 3 flow low-low	2oo3	64.5 t/h	25.8 t/h	2 s
FZLL-2004	SIF-201	H-201 pass 4 flow low-low	2oo3	64.5 t/h	25.8 t/h	2 s
PZLL-2009	SIF-202	H-201 burner FG pressure low-low	2oo3	1.0-2.0 barg	0.15 barg	2 s
BZLL-2008	SIF-202	H-201 loss of flame (all burners)	per burner / all	flame on	flame off	4 s (FFRT)
LZHH-2024	SIF-203	C-201 bottom level high-high	1oo1	50 %	85 %	10 s
HS-1121	SIF-204	P-112 area fire - manual / F&G confirmed	1oo2	-	-	30 s (valve)
HS-2041	SIF-204	P-204 area fire - manual / F&G confirmed	1oo2	-	-	30 s (valve)
HS-9000	SIF-901	Unit ESD-1 push-button	1oo2	-	-	2 s

6. SRS requirements (FEED)

- SIS sensors, logic solver and final elements independent of the BPCS (separate taps, transmitters, cables, JBs and cabinets); BPCS valves are not credited as SIS final elements.
- 2oo3 voting for continuous analogue initiators on heaters (degrade to 2oo2 on fault, 1oo1 trip on two faults); 1oo1 for SIL 1 level / pressure functions with BPCS-transmitter comparison alarm.
- De-energise to trip for all SIFs; line monitoring on SOV outputs; fail-safe on loss of power or air.
- Bypasses for maintenance only via key-switch with time limit and CCR alarm; operational overrides (start-up) auto-reset.
- Proof-test intervals: SIL 2 12 months (partial-stroke tests of SSOVs every 3 months), SIL 1 24 months; PFDavg to be verified with vendor data (IEC 61508 certified devices or prior use).
- Response times per SIF (table above) include sensor, logic solver and final element closure; SSOV closure $\leq$ 2 s, ROSOV $\leq$ 30 s.
- BMS: NFPA 85 / 86 and API 556 sequences (purge, pilot proving, flame supervision, safety time 4 s).

7. F&G executive actions

Confirmed gas (2ooN at 50 % LEL) or confirmed fire per fire zone initiate the actions shown in the C&E matrix: beacons / PAGA, isolation of fuel to the heaters in the heater zone, ROSOV closure for the hot pumps P-112 / P-204 in their zones and the signal to the OSBL fire-water pumps. F&G is a SIL 2 system; performance targets (coverage, detector voting) from the F&G mapping study.